

A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract—A detector scoring 99% on a machine-generated-text benchmark may have learned how machine language differs from human language, or how one corpus was punctuated, and the headline number does not separate the two. We decompose the signal into three arms. A surface-only model reads 47 orthographic features and never sees a word, a content-only model reads text stripped of punctuation, casing and non-ASCII characters, and a fine-tuned transformer on raw text is the reference. On HC3 the two arms are indistinguishable under a paired test, 3.20% against 3.26% error, and the null holds on five of five partitions with the sign changing between them. DeBERTa reaches 0.28% on the same split, so the parity joins two weak arms rather than a ceiling. On DAIGT V2 content is stronger by a factor of 7.9 in error. Subgroup splits localise that result rather than confirm it. The surface arm reaches 0.01% error on the Reddit sub-corpus, 74.8% of HC3, and loses to content on both other domains large enough to test. Removing the cue responsible, the rate of spaces before punctuation, costs that arm 10.03 points. BERT emits identical token identifiers for strings differing only by that cue, cannot represent it, and still reaches 0.9916 weighted F1. Surface form is informative on both corpora. We claim only that it reaches parity with content on one, and there for one reason.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3]. Read directly, those numbers say the task is close to solved. Read carefully, they say something weaker, because a benchmark score measures the separability of a particular corpus and not the capability a reader infers from it.

Separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second are documented, including a whitespace convention in HC3 [4], a length confound in the same corpus [5] and prompt-specific collection shortcuts [6]. What is missing is a way to ask how much, in total, a benchmark’s separability owes to surface form rather than content, and to place two benchmarks on the same axis.

This paper proposes that measurement. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters are stripped. A *full* model is a fine-tuned transformer on raw

text. The first two share a classifier family and are directly comparable, and the third is a reference point rather than a matched arm. The object of measurement throughout is a corpus and not a detector. Every model here is an instrument for reading a benchmark, and we offer no detector comparison, since a fair one would need matched training data on both corpora.

Applied to DAIGT V2 and HC3, the answer differs sharply. On HC3 the two arms are indistinguishable, 3.20% against 3.26% test error, while on DAIGT V2 content is stronger by a factor of 7.9. Splitting each corpus by its own subgroup labels then shows the HC3 tie is not a property of the benchmark. It belongs to the Reddit sub-corpus that is three quarters of it, and reverses on the other domains.

Our contributions are three. The first is the decomposition, stated formally in Section III-C as a reusable measurement over any human-versus-machine corpus, with the length control that stops its two arms sharing a channel. The second is its application to two benchmarks and their eighteen sub-corpora, which locates the HC3 result in one collection convention in one dominant domain and shows that removing that convention costs the surface arm ten points. The third is a tokenisation control establishing that the most-discussed cue in HC3 is sufficient in isolation yet unnecessary in practice, since BERT provably cannot represent it and still reaches 0.9916 weighted F1 there.

II. RELATED WORK

Detection on these benchmarks. Guo et al. [1] introduce HC3 with a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since, though Su et al. [2] show that regime narrows to roughly 91.7% under semantic-invariant tasks. Zhou [3] reaches competitive DAIGT V2 accuracy with character n-grams and no transformer, a result consistent with ours but reported without the surface-content separation that would explain it. Yadagiri et al. [7] pair DeBERTa with stylometric features and Socolof et al. [8] detect with style embeddings alone, both combining the two channels to raise a score where we hold them apart to measure one.

The closest prior result. Tian et al. [4] identify the HC3 whitespace convention, release a cleaning kit, and go further than they are usually credited for. Their appendix reports that a detector consisting of a single test for one token identifier reaches 82.12 F1 on HC3 at sentence level, above the

81.89 they quote for a fine-tuned RoBERTa. Our document-level equivalent of that rule reaches 94.22 weighted F1, and the difference is sentence against document rather than a disagreement. Theirs is the closest existing statement of this paper’s HC3 finding. What we add is a matched content arm to weigh the cue against, a per-domain split that locates it, and a tokenisation control asking whether a given detector can represent the cue at all.

Shared benchmarks and shortcut learning. RAID [9], M4 [10] and SemEval-2024 Task 8 [11] make detectors comparable across generators, domains and attacks. They do not ask what any one condition is separable by, which is the axis added here and the reason this measurement would apply to them unchanged. The pattern is not specific to text detection. Torralba and Efros [12] showed vision datasets are identifiable from their own images, and in natural language inference a model reading only the hypothesis reaches far above chance [13], [14], with Geirhos et al. [15] giving the general account. Our surface arm is a partial-input baseline transferred to this task, except that it removes a property of the text rather than a span of it. Feng et al. [16] give the caveat that governs how such a baseline may be read. A partial-input model scoring highly shows a dataset is cheatable, while one scoring poorly does not show it is clean, since the restricted representation may fail to express an artefact that is present. We therefore report the surface arm as an upper bound on surface-carried separability rather than as a measurement of it.

Robustness and fairness. Antoun et al. [17] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy. Liang et al. [18] show deployed detectors misclassify non-native English writing as machine-generated at high rates, which bears on ours, since a detector whose separability rests on orthographic regularity should concentrate its errors on writers whose orthographic habits differ from the corpus it was fitted to.

III. METHOD

A. Corpora and partitioning

DAIGT V2 [19] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems, assembled for a public detection competition. HC3 [1] contains 85,449 question-answer rows from five English sources, contrasting human answers with GPT-3.5-Turbo. The two make a useful pair because they differ in nearly every respect that matters here. DAIGT V2 has many generators, one genre and long documents, HC3 one generator, five source domains and short ones. Both were class-balanced by downsampling, giving 34,994 and 53,806 rows, which fixes a degenerate single-class predictor at 0.333 weighted F1 rather than something respectable.

HC3 carries 6,118 duplicate rows, 7.16% of the corpus, so the split is group-aware. Documents are grouped by an MD5 hash of their whitespace-normalised, lowercased text and whole groups go to one partition. The partition is 72/8/20, since a fifth goes to test first and a tenth of the remainder to

validation. Measured directly, the group-aware rule leaks 0 of 10,732 HC3 test documents where a plain stratified split of the same sample leaks 570, or 5.30%. The split is built once and reused by every model and seed.

B. Models

Five families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each under bag-of-words and TF-IDF. Two are transformers, `bert-base-uncased` [20] and `microsoft/deberta-v3-base` [21], fine-tuned over a sixteen-run grid per dataset, eight per model, varying learning rate, batch size and weight decay, with the operating point selected on validation weighted F1. DeBERTa’s SentencePiece tokeniser [22] encodes leading whitespace whereas BERT’s WordPiece does not, which makes the pair a controlled contrast on exactly the cue Section IV-D examines.

C. The decomposition

Write x for a document and $y \in \{0, 1\}$ for its label, with 1 denoting machine-generated. The surface map ϕ_S sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics covering punctuation rates, whitespace behaviour including spaces before punctuation, casing, document and sentence length, non-ASCII and digit rates. It reads no word identity. The content map ϕ_C lowercases, replaces every character outside the lowercase Latin alphabet and whitespace with a space, collapses runs of whitespace, then takes raw bag-of-words counts, so punctuation, casing and non-ASCII cannot enter it.

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg \min_h \sum_{(x,y) \in \mathcal{D}_t} \ell(h(\phi_a(x)), y) + \frac{1}{2C} \|h\|_2^2, \quad (1)$$

for $a \in \{S, C\}$, with C fixed at the library default for both. Sharing the classifier family and regularisation is what makes the two arms comparable, and Section IV-B reports what changes when C is chosen on validation for each arm instead. We report the held-out error ε_a of each arm and the gap $\Delta = \varepsilon_S - \varepsilon_C$. A gap near zero says orthography alone reaches the accuracy of content alone. The comparison is between two specific arms rather than a partition of the total separability, so a richer featurisation would lower either side, and it is not a claim that a corpus is clean.

One channel reaches both arms, since the surface arm reads size counts directly and raw bag-of-words rows sum to document length. Length is a documented confound on both corpora [5], so we close it, dropping the five size-scaling surface features and normalising content rows to unit total before rescaling by the mean training document length. The rescaling matters, since rows summing to one shrink every feature and the arm would otherwise collapse for reasons of scale rather than length.

TABLE I

TEST ERROR FOR ALL EIGHT CONFIGURATIONS. TRANSFORMER ROWS ARE DEPLOYED CHECKPOINTS, NEVER GRID MAXIMA. BOLD MARKS THE BEST ENTRY PER CORPUS AND UNDERLINE THE BEST CLASSICAL ONE.

Model	Rep.	DAIGT V2 err %	HC3 err %
Naive Bayes	BoW	4.09	12.87
Naive Bayes	TF-IDF	4.23	13.28
Logistic Reg.	BoW	1.07	<u>4.49</u>
Logistic Reg.	TF-IDF	1.43	6.35
SVM	BoW	1.30	5.25
SVM	TF-IDF	<u>0.90</u>	5.51
BERT	raw	0.84	0.84
DeBERTa	raw	0.83	0.28

D. Statistics and controls

Transformer baselines are read from the deployed checkpoint’s own record rather than from a grid maximum. Where two models are compared on the same test set the comparison is paired. Writing b and c for the discordant counts, we report the exact two-sided binomial p -value [23] together with a paired bootstrap 95% interval on the error difference over 10,000 resamples [24], resampling document indices so the pairing is preserved. A bootstrap says nothing about how the corpus was partitioned, and the central claim here is a null, so the decomposition is repeated over five group-aware partitions varying only the split seed. We run no parametric tests at $n \leq 30$.

Two controls guard the claims below. A *tokenisation* control compares token identifier sequences for minimally different strings, since whether a detector can exploit a character-level cue is a property of its tokeniser and directly testable. A *pipeline-fires* control confirms that a cleaning step reporting no change actually executed, by applying the same pipeline to the corpus where it removes something the tokeniser can represent.

E. Reproducibility

Partitions are built once at split seed 42 and reused by every model and every training seed, so only initialisation and batch order vary between runs. Re-splitting per seed would change the evaluation set between runs and invalidate the comparison. The four deployed checkpoints were retrained at seeds 123 and 456 for the ranges reported in Section IV. Fine-tuning ran three epochs at a 128-token budget on one 8 GiB consumer GPU, the longest run taking 1,111 seconds at 4.22 GiB peak memory, under PyTorch 2.11, Transformers 4.57 and scikit-learn 1.9. Code, split manifests and the per-run record behind every number reported here are kept in a public repository, withheld here for anonymity.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 until the text budget is matched

Table I reports every configuration, and the two benchmarks part company. On DAIGT V2 the best classical configuration sits 0.07 error points above the best transformer, and on the same documents the two disagree on 99 cases split 47 to 52, so McNemar returns $p = 0.69$ and the paired interval on the error

TABLE II

TOP, THE DECOMPOSITION ON EACH CORPUS, WITH THE LOWER ROWS CLOSING THE DOCUMENT-LENGTH CHANNEL BOTH ARMS SHARE. BOTTOM, HC3 BY SOURCE DOMAIN, EACH REBALANCED TO PARITY, WITH THE SHARE OF DOCUMENTS CARRYING A SPACE BEFORE PUNCTUATION.

Arm	DAIGT V2 err %		HC3 err %	
surface-only	7.86		3.20	
content-only	0.99		3.26	
full transformer	0.83		0.28	
length-only (3 feats)	24.59		15.23	
surface-only, no length	9.93		3.37	
content-only, length-norm.	0.94		6.28	

HC3 domain	n	surface		content		cue present in	
		n	err %	n	err %	human	machine
reddit_eli5	6,690	0.01	2.66	98.9%	0.3%		
finance	684	7.60	3.07	5.6%	0.1%		
medicine	250	4.00	0.40	17.0%	0.1%		
open_qa	198	4.04	5.05	60.7%	0.4%		
wiki_csai	168	5.36	11.90	1.6%	0.2%		

open_qa and wiki_csai fall below 200 test rows.

difference, $[-0.21, +0.34]$ points, contains zero. On HC3 the same comparison gives 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of +4.21 points.

Repeating the four deployed checkpoints at seeds 123 and 456 moves them very little. DAIGT V2 BERT averages 0.9921 over a range of 0.0011 and DeBERTa 0.9930 over 0.0024, so the two overlap. HC3 BERT averages 0.9930 over 0.0036 and DeBERTa 0.9970 over 0.0005, and those ranges are disjoint. The tie and the separation both survive reseeding, in the directions the paper needs.

That tie is also not a matched comparison. The classical models read whole documents while the transformers read 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer. Refitting the classical grid on the exact character span each checkpoint’s tokeniser kept, taken from offset mapping with no decode round trip, moves the best DAIGT V2 configuration from 0.90% to 2.10% error against BERT’s 0.84%, a difference of 1.26 points with interval $[+0.90, +1.63]$, and the DeBERTa window returns the same difference. On HC3 the same refit widens the gap from 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora, so the unmatched tie was a fact about information access rather than about architecture.

B. On HC3, orthography alone matches content alone

Table II gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content, disagreeing on 625 documents split 316 to 309, so McNemar returns $p = 0.81$ and the bootstrap puts the difference at -0.07 points with interval $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary. On DAIGT V2 the same arms separate by a factor of 7.9, 7.86% against 0.99%, with 569 discordant documents split 44 to 525 at $p < 10^{-6}$.

The content arm is deliberately unfiltered. Refitting it with the stopword removal and lemmatisation Table I’s classical pipeline reproduces that pipeline exactly, and on HC3 the filtered arm then loses to orthography by 1.30 points where the unfiltered arm ties it. We report the unfiltered arm because it is the stronger opponent, which makes the parity claim harder rather than easier, and against it the HC3 classical-to-transformer error ratio is 11.6 to one rather than the sixteen to one Table I implies.

Surface form is informative on both benchmarks, since DAIGT V2’s surface arm reaches 0.9214 weighted F1. The finding is not that one corpus is clean, but that only on HC3 does surface rise to parity with content. Closing the length channel sharpens both conclusions rather than weakening them, widening DAIGT V2’s content advantage from 7.9 to 10.6 times and turning HC3’s parity into a 2.91-point advantage for orthography. A null on one partition is what a lucky split manufactures, so the decomposition runs again over five group-aware partitions. Content leads on DAIGT V2 on all five at $p < 10^{-6}$. On HC3 the difference reaches significance on none, with p of 0.81, 0.27, 0.34, 0.23 and 0.78, and it changes sign between partitions, which is the strong form of the null since an underpowered real difference keeps its sign.

Both arms take the library default inverse regularisation strength, which makes them comparable to each other rather than to their own best, and they are not symmetric in dimension. Choosing that value on validation for each arm separately selects 100 for surface and 0.1 for content on both corpora and changes no conclusion. On HC3 the two arms move to 3.15% and 3.47% error and remain indistinguishable on all five partitions, at p from 0.19 to 0.75 with every interval containing zero. The surface arm is then nominally ahead on all five by 0.08 to 0.32 points, so the sign argument above belongs to the default fit and the tuned fit rests on the intervals alone. On DAIGT V2 content keeps its advantage on all five at $p < 10^{-6}$, 0.96% against 7.66% on the main partition.

C. The parity belongs to one sub-domain and one cue

Both corpora carry subgroup labels, and rejoining them to the existing splits recovers a source for every row. The lower half of Table II runs the same measurement per HC3 domain. On `reddit_eli5` the surface arm reaches 0.01% error, roughly one mistake in 6,690 documents. On the two other domains with enough test rows to support a claim the ordering reverses and content wins, by 4.53 points on `finance` with interval $[+2.34, +6.87]$ and by 3.60 on `medicine` with interval $[+1.60, +6.00]$, neither containing zero. Since `reddit_eli5` is 74.8% of the balanced corpus before the per-domain rebalancing this table applies, the corpus-level parity is substantially that one domain.

The mechanism is measured rather than inferred. The space-before-punctuation cue appears in 98.9% of `reddit_eli5` human documents against 0.3% of machine ones, and decays across the rest to 1.6% on `wiki_csai`. A single boolean rule, treating a document as human when the cue is present, scores 99.22% accuracy on `reddit_eli5` and 94.23% on

the balanced corpus at document level, 94.22 weighted F1, against the 82.12 F1 Tian et al. [4] report for the sentence-level equivalent. Applying their cleaning kit, and nothing else, changes 44.5% of HC3 documents and moves the surface arm from 3.20% to 13.22% error, a loss of 10.03 points with interval $[-10.69, -9.38]$, while the content arm is untouched at 3.26%. After cleaning, orthography sits 9.96 points behind content on the corpus where it had tied.

The same treatment on DAIGT V2 pairs each machine generator against the shared human pool. Content wins on every generator, but the surface arm’s error spans sixteenfold across the ten generators with enough test rows to support a claim, from 0.50% to 8.13%. Two pairs of the same underlying model contributed by different people differ by factors of 4.1 and 2.3, and we report those two as suggestive rather than established, since the smaller member of each pair holds 184 and 130 test rows and falls below the 200-row floor this paper uses elsewhere. Surface separability there tracks collection provenance more closely than model identity, which is the conclusion the HC3 domain split reaches by another route.

D. A model that cannot represent the cue reaches 0.9916 anyway

The natural inference from the cue’s dominance is that HC3-trained detectors exploit it. Tokenisation makes that testable, and the test does not support it. BERT’s WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace, emitting identical identifier sequences for "the answer is simple ." and "the answer is simple." in three of three pairs tested, while DeBERTa’s SentencePiece distinguishes all three. BERT cannot represent the cue and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice.

This is also why removing the cue changes nothing for BERT. Whitespace cleaning on HC3 yields predictions bit-identical to the uncleaned run, which alone would be indistinguishable from a cleaning step that never executed. The same pipeline on DAIGT V2, where it also removes emoji that WordPiece represents, does change BERT’s predictions, so the pipeline fires and the null is specific to whitespace. We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue, since the two models differ in tokeniser, architecture, pretraining corpus and parameter count. What the experiment establishes is that a null from removing a cue is uninterpretable without first checking the model could read it. Fig. 1 collects the results above.

E. Neither corpus transfers to the other

Each deployed checkpoint was also evaluated on the corpus it was not trained on, without adaptation, over three seeds. Every cell loses between 0.0833 and 0.2019 mean weighted F1. Trained on DAIGT V2, BERT falls from 0.9921 in domain to 0.7902 over a range of 0.770 to 0.810 and DeBERTa from 0.9930 to 0.9096 over 0.891 to 0.926. Trained on HC3, BERT falls from 0.9930 to 0.8311 over 0.813 to 0.852 and DeBERTa

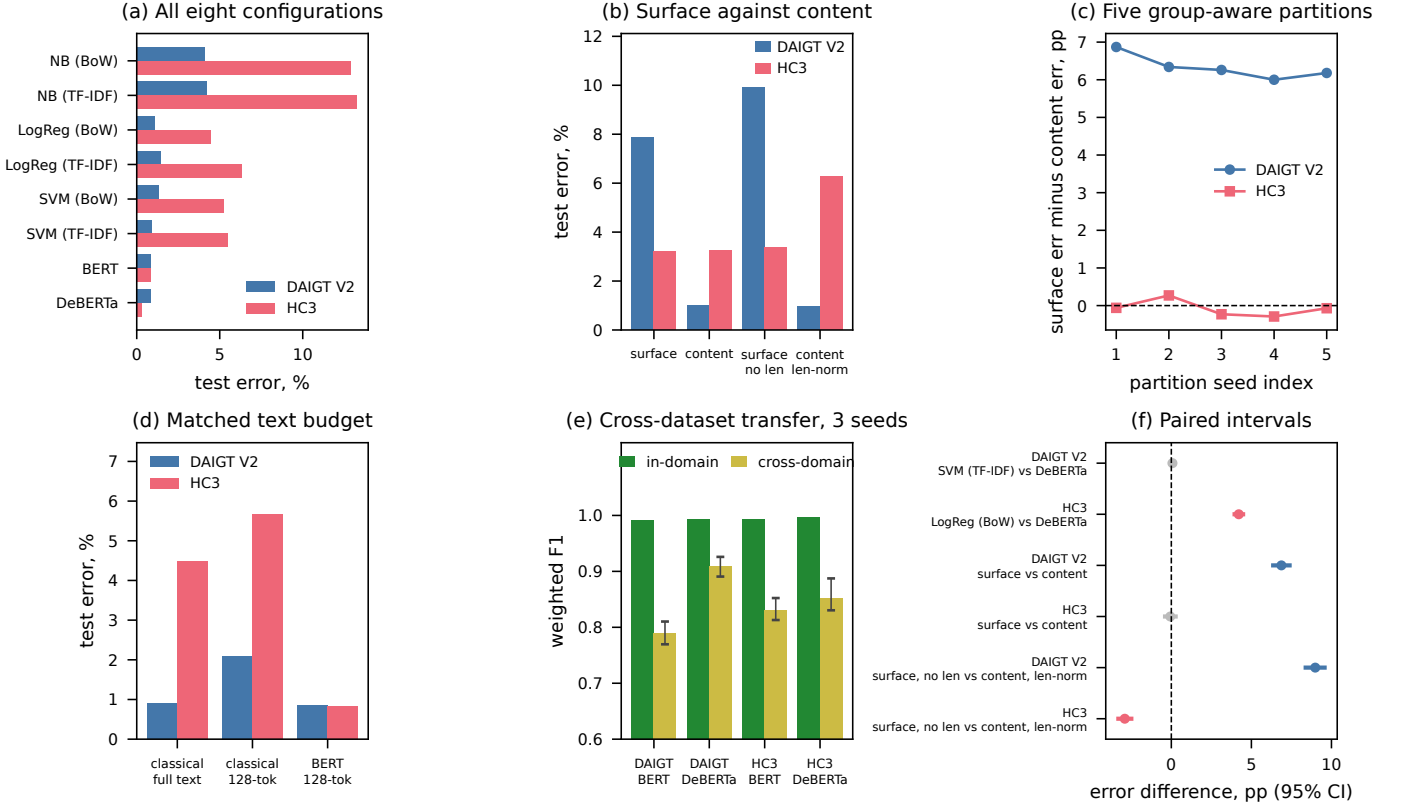


Fig. 1. Every headline result in one view. Panels give the model ranking (a), the decomposition with and without the length channel (b), the surface-minus-content difference across five partitions (c), the matched text budget (d), cross-corpus transfer (e), and the paired intervals behind each significance claim (f), where grey marks an interval containing zero.

from 0.9970 to 0.8512 over 0.830 to 0.887. DeBERTa’s mean gap is roughly half BERT’s in both directions, which we report as a mean-level pattern rather than a confirmed effect, since the observed ranges for BERT’s two directions nearly touch and three runs do not support a test. A substantial part of what each model learned is specific to the corpus it was fitted to, which is what the decomposition anticipates when much of a corpus’s separability is carried by how that corpus was assembled.

F. A published detector out of domain

Every model above is one we trained. Hello-SimpleAI’s RoBERTa detector, released with HC3 by that corpus’s authors [25], gives an external reference point, and we ran it over the same test partitions without fine-tuning. On HC3 it reaches 0.9952 weighted F1, which is contaminated, since the detector was trained on that corpus and our test rows were almost certainly in its training data. On DAIGT V2, which it has never seen, it reaches 0.8230 weighted F1 at 17.55% error, against 0.83% for our DeBERTa checkpoint trained in domain. One model was trained on the corpus being scored and the other was not, so the gap is not an architecture result, and the published detector reads 512 tokens against our 128, which if anything favours it. We ran it at its own decision boundary with no recalibration, so the figure bounds how far the 99.82 F1 quoted in Section II travels off its own corpus untuned rather than in principle.

V. DISCUSSION AND CONCLUSION

The decomposition separates two benchmarks that a headline accuracy figure does not, and applied per sub-corpus it separates one benchmark from itself. On HC3 orthography alone matches content alone and a paired test cannot distinguish them on any of five partitions, while on DAIGT V2 content is stronger by a factor of 7.9. Section IV-C then narrows the first of those to one collection convention in the Reddit sub-corpus that is three quarters of the corpus, and shows that removing that convention costs the surface arm ten points. A single collection convention can be cleaned. The diffuse stylistic signal DAIGT V2’s surface arm reads cannot, and may not even be an artefact.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2’s surface arm reaches 0.9214 weighted F1, so surface form is substantially informative on both, and Feng et al. [16] give the reason a low surface score would not have settled the question either. The finding is comparative and bounded by the arms we built.

Two results here say more about method than about these corpora. A cleaning experiment reporting no change is uninterpretable without evidence the pipeline ran, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than the corpus. Both requirements cost one forward pass and neither is standard practice.

Several limits bound every number above. The transformers see at most 128 tokens, so the DAIGT V2 transformer results

describe classification from an essay’s opening, which is why the classical comparison is repeated on that same span in Section IV. The four deployed checkpoints were run at three seeds and move by 0.0005 to 0.0036 weighted F1, but the grid behind them is single-seed. The content arm is bag-of-words, so word order and syntax lie outside it, and the surface set is 47 hand-built features, so ε_S is an upper bound. Five of the eighteen sub-corpora fall below 200 test rows after rebalancing and are reported as underpowered rather than as nulls. HC3 has one generator collected at one time and both corpora are English, so the orthographic conventions the surface arm reads are not language-independent.

The measurement itself is the portable part. It needs no new annotation, it fits two logistic regressions, and it runs in minutes. It yields one number per corpus, and per sub-corpus, that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, in the same way a hypothesis-only baseline is now reported alongside a natural language inference dataset [14].

REFERENCES

- [1] B. Guo, X. Zhang, Z. Wang, M. Jiang, J. Nie, Y. Ding, J. Yue, and Y. Wu, “How close is chatgpt to human experts? comparison corpus, evaluation, and detection,” arXiv, 2023, preprint, arXiv:2301.07597.
- [2] Z. Su, X. Wu, W. Zhou, G. Ma, and S. Hu, “Hc3 plus: A semantic-invariant human chatgpt comparison corpus,” arXiv, 2023, preprint, arXiv:2309.02731.
- [3] C. Zhou, “Exploiting machine learning model ensemble for AI-generated texts detection,” *Transactions on Computer Science and Intelligent Systems Research*, vol. 5, 2024, aIDML 2024. [Online]. Available: <https://wepub.org/index.php/TCSISR/article/view/2382>
- [4] Y. Tian, H. Chen, X. Wang, Z. Bai, Q. Zhang, R. Li, C. Xu, and Y. Wang, “Multiscale positive-unlabeled detection of AI-generated texts,” arXiv, 2023, preprint, arXiv:2305.18149.
- [5] M. S. Baidya, S. S. Baidya, and C. Chawla, “Detecting the machine: A comprehensive benchmark of AI-generated text detectors across architectures, domains, and adversarial conditions,” arXiv, 2026, preprint, arXiv:2603.17522.
- [6] C. Park, H. J. Kim, J. Kim, Y. Kim, T. Kim, H. Cho, H. Jo, S. goo Lee, and K. M. Yoo, “Investigating the influence of prompt-specific shortcuts in AI generated text detection,” arXiv, 2024, preprint, arXiv:2406.16275.
- [7] A. Yadagiri, S. Sai Teja, P. Pakray, and C. Chunka, “AI-generated text detection using DeBERTa with auxiliary stylometric features,” in *Proceedings of the RANLP 2025 Workshop on Multi-Domain Detection of AI-Generated Text (M-DAIGT)*, 2025. [Online]. Available: <https://aclanthology.org/2025.ranlp-mdaiht.2/>
- [8] G. Z. Socolof and R. Kacholia, “Fast, interpretable AI-generated text detection using style embeddings,” Stanford CS224N final project, 2024, not peer-reviewed. [Online]. Available: <https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1244/final-projects/GiuliaZoeSocolofRitikaKacholia.pdf>
- [9] L. Dugan, A. Hwang, F. Trhlík, A. Zhu, J. M. Ludan, H. Xu, D. Ippolito, and C. Callison-Burch, “RAID: A shared benchmark for robust evaluation of machine-generated text detectors,” in *Proc. ACL*, 2024, pp. 12463–12492.
- [10] Y. Wang, J. Mansurov, P. Ivanov, J. Su, A. Shelmanov, A. Tsvigun, C. Whitehouse, O. Mohammed Afzal, T. Mahmoud, T. Sasaki, T. Arnold, A. F. Aji, N. Habash, I. Gurevych, and P. Nakov, “M4: Multi-generator, multi-domain, and multi-lingual black-box machine-generated text detection,” in *Proc. EACL*, 2024, pp. 1369–1407.
- [11] Y. Wang, J. Mansurov, P. Ivanov, J. Su, A. Shelmanov, A. Tsvigun, O. M. Afzal, T. Mahmoud, G. Puccetti, T. Arnold, A. F. Aji, N. Habash, I. Gurevych, and P. Nakov, “SemEval-2024 task 8: Multidomain, multi-model and multilingual machine-generated text detection,” in *Proc. 18th Int. Workshop on Semantic Evaluation (SemEval-2024)*, 2024, pp. 2057–2079.
- [12] A. Torralba and A. A. Efros, “Unbiased look at dataset bias,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2011, pp. 1521–1528.
- [13] S. Gururangan, S. Swayamdipta, O. Levy, R. Schwartz, S. R. Bowman, and N. A. Smith, “Annotation artifacts in natural language inference data,” in *Proc. NAACL-HLT*, 2018, pp. 107–112.
- [14] A. Poliak, J. Naradowsky, A. Haldar, R. Rudinger, and B. Van Durme, “Hypothesis only baselines in natural language inference,” in *Proc. Joint Conf. on Lexical and Computational Semantics (*SEM)*, 2018, pp. 180–191.
- [15] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.
- [16] S. Feng, E. Wallace, and J. Boyd-Graber, “Misleading failures of partial-input baselines,” in *Proc. ACL*, 2019, pp. 5533–5538.
- [17] W. Antoun, V. Moulleron, B. Sagot, and D. Seddah, “Towards a robust detection of language model generated text: Is chatgpt that easy to detect?” arXiv, 2023, preprint, arXiv:2306.05871.
- [18] W. Liang, M. Yuksekgonul, Y. Mao, E. Wu, and J. Zou, “GPT detectors are biased against non-native English writers,” *Patterns*, vol. 4, no. 7, p. 100779, 2023.
- [19] thedrcat, “DAIGT v2 train dataset,” Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/thedrcat/daigt-v2-train-dataset>
- [20] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [21] P. He, J. Gao, and W. Chen, “DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023, arXiv:2111.09543.
- [22] T. Kudo and J. Richardson, “SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing,” in *Proc. EMNLP: System Demonstrations*, 2018, pp. 66–71.
- [23] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
- [24] B. Efron, “Bootstrap methods: Another look at the jackknife,” *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [25] “ChatGPT detector, RoBERTa,” Hugging Face Models, 2023. [Online]. Available: <https://huggingface.co/Hello-SimpleAI/chatgpt-detector-roberta>